

DEMOGRAPHIC RESPONSES OF A TROPICAL CLADOCERAN TO CADMIUM: EFFECTS OF FOOD SUPPLY AND DENSITY

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Abstract. Life table response experiments (LTRE) were performed to investigate effects of food supply on density-dependent effects on life history and population responses of the tropical cladoceran *Moinodaphnia macleayi* to cadmium. Its short life cycle and relatively constant age-specific birth and mortality rates allowed population growth rates (λ) to be accurately estimated from short life table experiments and by using a simplified two-stage demographic model. Decomposition and regression analyses of λ showed that density-dependent effects on population responses to cadmium were modified by food availability through density-dependent effects on mortality and reproductive rate. At moderate food levels (1.8 $\mu\text{g C/mL}$ of *Chorella vulgaris*) and low densities (<125 animals/L), the effects of cadmium on λ decreased with increasing population density, due to antagonistic effects between the contributions of age at first reproduction and daily reproduction on λ . At high densities (>125 animals/L), the negative effects of cadmium on daily reproduction rates increased with density; hence density promoted the negative effects of cadmium on λ . At low food levels (0.4 $\mu\text{g C/mL}$ of *Chorella vulgaris*), increasing population density reduced juvenile survival, increasing food per head. Thus adverse effects of cadmium on reproduction, which had the greatest contribution to λ , were buffered by increasing population density. Regression analysis performed on population responses across increasing population density levels and cadmium concentrations showed that at high densities and low food levels ecological compensation will prevent populations at the steady-state equilibrium size from being driven to extinction by toxicity effects at the individual level. Alternatively, at low densities, when food availability is not limiting, exposure to toxic substances can increase extinction probability. These results indicate that risk assessment procedures based on demographic analysis performed at low densities and high food levels may overestimate the ecological risk posed by toxic substances.

Key words: cadmium; *Daphnia*; density-dependent effects; food availability; life table response experiments (LTRE); *Moinodaphnia macleayi*; population growth rate.

INTRODUCTION

In chemical risk assessment, the population growth rate (measured either as the multiplication rate per unit time, λ , or as the Malthusian parameter, $r = \ln(\lambda)$) has been proposed as a more relevant measurement of toxicant effects than traditional measures of mortality or reproduction. This is argued from the viewpoint that population growth rate integrates life history responses, providing measures of impact at higher levels of ecological organization (Sibly 1996, Calow et al. 1997, Forbes and Calow 1999). To make predictions concerning an individual's contribution to population growth, age at reproduction, offspring production, and juvenile and adult survival, probabilities are measured and combined to calculate predicted population growth rates (Calow et al. 1997). Life table response experiments (LTRE) have made a valuable contribution to

the assessment of population consequences of toxicant effects on individual life history traits (Caswell 1996, Levin et al. 1996). In LTRE, the extent to which population growth rate responds to toxicant exposure is determined by analyzing the severity of toxic effects on individual life history traits as well as the sensitivity of λ to changes in individual traits contributing to it (Caswell 1996, Levin et al. 1996). Nevertheless, most experiments examining population response to toxic substances have been carried out at low density and/or high food levels. In general, increasing densities and reducing food availability increase competition for food and other resources, so individuals get less food, have fewer resources available for defense against environmental stress, and hence are more susceptible to toxic stress (Sibly 1999). Thus, laboratory experiments can severely underestimate population responses to toxic stress at typical densities and food levels (Grant 1998, Sibly 1999). Nevertheless, simulation studies predict that when mortality or production rates are affected by population density, more food is available for the survivors and individuals therefore have more resources to cope with stress. Thus, negative toxic ef-

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fects on population dynamics may be buffered under conditions of high population density (Sibly 1996, Calow et al. 1997, Grant 1998).

Experimental studies performed to assess the effects of population density or food availability on population responses to toxicants have found that negative toxic effects on population dynamics are ameliorated under conditions of high population density and moderate toxicant exposure levels (Postma et al. 1994, Linke-Gamenick et al. 1999) or under very limiting food conditions (Koivisto et al. 1992). However, alternative responses, where density or food-limiting conditions exacerbate or do not affect toxicity have also been reported (Ramakrishna-Rao and Sarma 1986, Chandini 1988, 1989, Klüttgen and Ratte 1994, Klüttgen et al. 1996, Linke-Gamenick et al. 1999). Nevertheless, except in the studies of Postma et al. (1994) and Koivisto et al. (1992), food in these experiments was not limiting, since mortality was only marginally affected under increasing population density or reducing food availability. These studies suggest that density-dependent effects on population responses to toxic substances can be modified by the availability of food through density-dependent effects on mortality and production rates. In a simulation study, Grant (1998) found that when juvenile survival was density dependent, toxic effects of dieldrin on population dynamics of the copepod *Eurytemora affinis* were ameliorated by increasing density. Conversely, the opposite trend was observed when fecundity was density dependent. Thus, it could be hypothesized that at barely limiting food conditions, when production rates (growth and reproduction) but not survivorship are affected by population density, negative toxic effects on population dynamics would be exacerbated by increasing density, due to a synergistic interaction between density and toxicity. Alternatively, at more limiting food conditions, when population density affects both juvenile and adult survival, density-dependent effects are likely to compensate for toxicant-induced reductions in population growth rates, since more food will be available to the survivors, compensating production rates (Sibly 1996).

Here we test this hypothesis by studying the combined effects of density and toxicity on life history and population responses of the tropical cladoceran *Moinodaphnia macleayi* under varying food levels. Cadmium was selected as a model toxic substance, since it has been previously shown to influence survival and reproductive parameters in cladoceran species at relevant field concentrations (Barata et al. 1999, Barata and Baird 2000). Cadmium is a heavy metal associated with many industrial activities. In particular, in northern Australia the presence of cadmium in freshwater above background levels (0.1–2 µg/L) is associated with fertilizers and mining activities (Noller et al. 1994, Markich and Brown 1998).

Moinodaphnia macleayi is a filter-feeding cladoc-

eran of body size 0.2–0.5 mm, which inhabits the floodplain of tropical northern Australia (Jilli 1986). On the basis of its success in laboratory culture and its sensitivity to toxic substances *M. macleayi* is used as a key species in the assessment of toxicity of waste waters from mining activities in tropical northern Australia (Hyne et al. 1993). Its short life cycle and high reproductive performance allows the determination of population growth rate parameters from short life history toxicity tests. Thus cost-effective complex experimental designs can be performed (Sibly 1999). Furthermore, individual genotypes can be maintained indefinitely, since females reproduce by ameiotic parthenogenesis in the laboratory. Thus, by using single genotypes, undesirable sources of variation such as genetic differences in the study populations can be controlled (Lynch 1989).

MATERIAL AND METHODS

Experimental animals

Individuals of a single clone of *M. macleayi* originally collected from Magela Creek, Northern Territory, Australia, were selected for this study. This particular clone was chosen because it is currently used in toxicity tests in Northern Australia in a number of universities, government and industrial laboratories (Sunderam et al. 1994, Rippon and Riley 1996).

Maintenance of bulk cultures

We maintained 10–20 bulk cultures of 10 animals in 150 mL of ASTM soft water (American Society for Testing and Materials 1998), in 180 mL screw top glass jars, with the addition of a standard organic extract (Baird et al. 1989). Animals were fed with *Chlorella vulgaris* (10⁶ cells/mL, corresponding to 3.6 µg C/mL sensu Barata and Baird [1998]), which was grown axenically in *Jaworski/Euglena gracilis* 1:1 medium (Culture Collection of Algae and Protozoa 1989). Photoperiod was set to 12 h light:12 h dark, and temperature retained at 26 ± 0.5°C. Cultures were changed every other day with freshly prepared medium and neonates removed within 24 h of release. Every week cultures were reset with newly released neonates.

Feeding responses

To test our hypothesis, we selected two food concentrations of *C. vulgaris*, 5 × 10⁵ and 10⁵ cells/mL, with an equivalent carbon content of 1.8 and 0.4 µg C/mL, respectively (Barata and Baird 1998); hereafter referred as “moderate” and “low” food levels. These food concentrations were chosen as represent “incipient limiting” and “very limiting” food concentrations in relation to the functional feeding response of *M. macleayi* (Fig. 1). Indeed at moderate and low food levels the feeding rate of females of *M. macleayi* was 50% and 1% of the maximal feeding rate, respectively. Previous studies conducted in cladoceran species have

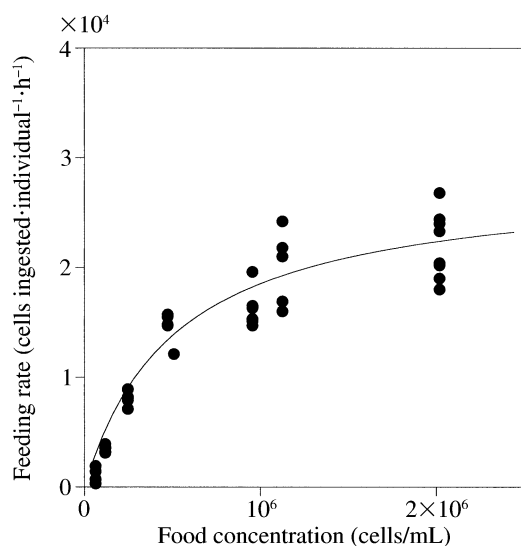


FIG. 1. Feeding rate responses of second brood females of *M. macleayi* to food concentration, with the fitted model as calculated from Eq. 1. $F_{\max} = 2.8 \times 10^4$ cells ingested·(individual) $^{-1}$ ·h $^{-1}$; $K = 52.1 \times 10^4$ cells/mL. The coefficient of determination of the model was 0.9. The size (mean ± 1 SD) of the females used was 0.49 ± 0.03 mm.

shown that only at very limiting food concentrations are juvenile and adult mortality rates significantly affected (Koivisto et al. 1992, McCauley et al. 1990). The functional feeding response of *M. macleayi* was determined by exposing groups of five 4–5-d-old females to eight different concentrations of *C. vulgaris*, ranking 0.5×10^5 – 20×10^5 cells/mL. Five replicates per concentration were used. At the start of the feeding experiments, groups of five animals were transferred to 10 mL borosilicate glass jars filled with 7 mL of ASTM soft water with the desired algae concentration. Individual feeding rates (cells·animal $^{-1}$ ·h $^{-1}$) were determined as the change in cell density during 24 h according to the method given by Allen et al. (1995). Cell density was measured using a Coulter Multisizer particle counter (Coulter Electronics, Luton, UK). Concentration feeding responses were fitted to the type 2 model of Holling (1966):

$$F = F_{\max} \frac{C}{C + K} \quad (1)$$

where F is feeding rate (cells ingested per individual per hour), $F_{\max}$ is the maximum feeding rate, C is cell density, and K the half-saturation food concentration constant (i.e., the concentration at which the feeding rate is half the maximal). This type of response has been described in different *Daphnia* species (Porter et al. 1982, Bohrer and Lampert 1988, Allen 1993).

Experimental design

Density-dependent effects on population responses of *M. macleayi* were studied at moderate and low food

levels in two separate experiments. Within each food concentration, 1170 neonates (<12-h-old) collected from the bulk cultures were randomly assigned to 150 replicate $\times$ population density $\times$ cadmium group combinations. Neonates originated from second-to-third brood females ($\sim$ 4–5-d-old) to minimize maternal effects (Hyne et al. 1993). Experimental treatment levels were designed in a logarithmic scale and consisted in six different cadmium concentrations (0, 0.25, 0.5, 1, 2, 4 μ g/L) and four population densities (cultures of 25, 125, 250, and 500 individuals/L). To simulate the four population densities, experimental animals were raised individually or in groups of 5, 10, or 20 in 40 mL of medium at a desired algal concentration in 50 mL borosilicate glass jars. To improve the accuracy of life history and population growth rate estimates, the treatments with densities of one individual were replicated 10 times, whereas the rest of treatments were replicated five times. Culture conditions were similar to those described for bulk cultures. Reproduction and survivorship of experimental animals were monitored every 12 h until first reproduction, after which they were monitored every day. Life table experiments lasted 10 d. During this period, most of the females deposited their first five clutches. Abbreviated life table experiments had shown that population growth rates (λ) of cladoceran populations cultured under these food conditions can be accurately determined from age at egg deposition and size of first clutches (Baird et al. 1990, Vanni and Lampert 1992, Mooij and Boersma 1996). The following demographic parameters were recorded: juvenile and adult survivorship; age of releasing neonates, which included age at first reproduction; and daily reproduction rates (number of neonates produced per surviving female per day).

Demographic analysis

Population growth rate (λ) in the different treatments was determined from the age-specific data on fecundities and survival probabilities using the five or 10 replicates. This value was computed iteratively from the simplified two-stage model of Sibly et al. (2000):

$$n(p_j t_j)(\lambda)^{-t_j} + p_a(\lambda)^{-1} = 1 \quad (2)$$

where n is the daily number of offspring produced per female, t_j is the mean time to first reproduction in days, p_j is the juvenile survival (proportion of juveniles surviving per day from birth to first reproduction), p_a adult survival rates (proportion of adults surviving per day averaged over the adult period). A time unit of one day was employed for all treatments because reproductive output was measured every day and since the experimental design required that animals were cultured in groups, thus preventing individual females from being easily identified. Estimates of 95% confidence levels of λ were obtained analytically as in Sibly et al. (2000). Although the use of age-dependent or age-classified models makes maximum use of the demographic in-

formation (Caswell 1996), for organisms with life histories that can be accurately described by age-independent birth, juvenile, and adult mortality rates, more accurate estimates of population growth rates can be obtained by using the two-stage model (Sibly et al. 2000). Furthermore, recent studies have shown that the assumptions that birth and mortality rates do not change with age may also be reasonable for organisms that do grow after the age of first breeding, as in the case of the polychaete *Capitella* (Linke-Gamenick et al. 1999, Sibly et al. 2000) and the copepod *Eurytemora affinis* (Sibly 1996). In our study the age-specific fecundity and probability schedules of *M. macleayi* females in the different treatments remained relatively constant during the experiments. As a result, λ estimates from the two-stage model in most cases were not significantly different from λ determined using an age-dependent model (95% CL of λ crosses the diagonal line that indicates identity; Fig. 2).

The effects of density and cadmium on life history traits (i.e., n , t_j , p_j , and p_a) do not necessarily translate into equivalent responses of λ . This will depend on the severity of the treatment's effects on the individual life history traits as well as the sensitivity of λ to changes of the individual traits that contribute to it. Thus, to determine by how much effects of density and cadmium on n , p_j , p_a , and t_j contribute to the overall impact on λ , a decomposition analysis of the contributions of each parameter was conducted as in Levin et al. (1996: Eq. 16) using the sensitivity equations given by Sibly et al. (2000: Eqs. 4–7). Differences between control (zero concentrations of cadmium and density of 25 daphnids/L) and treatments for each of the parameters and sensitivities were evaluated as the mean of the two parameter sets.

Cadmium analysis

Stock solutions of cadmium (expressed as total Cd concentration) were prepared by adding analytical reagent-grade salts (i.e., $3\text{CdSO}_4 \cdot 8\text{H}_2\text{O}$) to deionized water (18 M Ω /cm resistivity; Milli-Q, Millipore, Bedford, Massachusetts, USA). Nominal test concentrations of 0.25, 0.5, 1, 2, and 4 $\mu\text{g/L}$ of Cd were subsequently prepared by adding aliquots of each stock solution to the synthetic ASTM soft water. In each of the experiments, duplicate water samples (10 mL) were collected from experimental vessels in freshly prepared ($t = 0$ h) and old test solutions ($t = 48$ h). Water samples of Cd were acidified with concentrated nitric acid (Ultrapur, Merck, Dorset, UK). Actual concentrations of Cd were measured using an ATI Unicam Model 939-graphite furnace atomic absorption spectrometer (Unicam, Cambridge, UK) following the procedure described by Barata et al. (1998).

To ensure that toxicity was not affected by reduced levels of oxygen (van Leeuwen 1995), dissolved oxygen (DO) concentration levels were measured in fresh and old test solutions using a polarographic oxygen

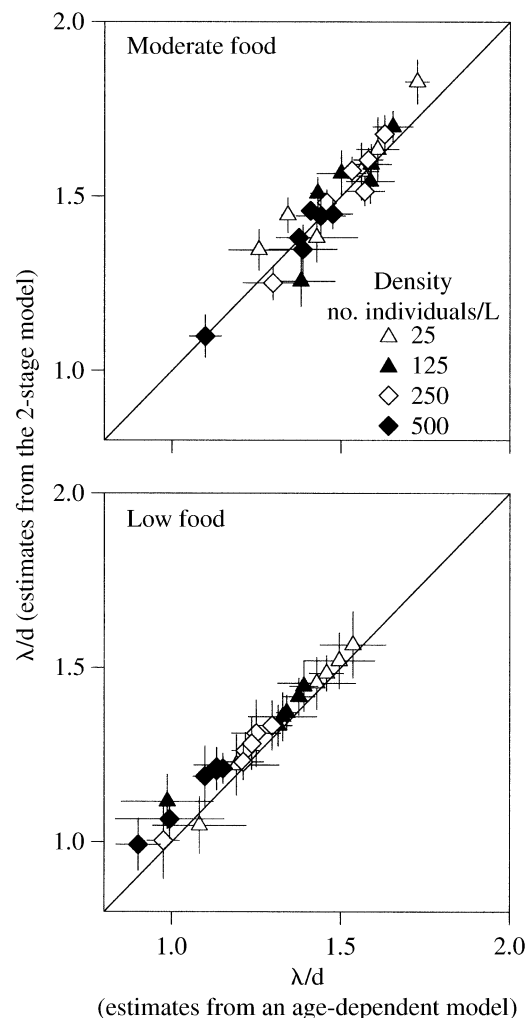


FIG. 2. Plots of mean $\lambda \pm 95\%$ CL estimates obtained from the two-stage model depicted in Eq. 2 (y-axis) vs. λ estimates determined from an age-dependent model (Meyer et al. 1986). Confidence levels of λ values obtained from the two-stage model were estimated analytically as in Sibly et al. (2000), whereas those from the age-dependent model were estimated from jackknife pseudovalues according to Meyer et al. (1986).

electrode coupled to a YSI model 57 oxygen meter (Yellow Springs Instruments, Yellow Springs, Ohio, USA).

Statistical analysis

At each of the studied food concentrations, the effects of population density and cadmium exposure on daily reproduction (ln-transformed) were tested by two-way ANOVA. Significant heteroscedasticity among treatments was observed ($P < 0.05$, Levene's test) due to the high variance levels of those treatments including population densities of 25 individuals/L, even after transformation. Therefore, F ratios were corrected according to Brown and Forsythe (1974). The effects of

TABLE 1. Measured vs. nominal concentrations ($\mu\text{g/L}$) of cadmium of freshly prepared (t_0) and old test solutions (t_{48} ; i.e., 48-h-old).

Nominal concentration	$N^\dagger$	Measured concentration ‡	
		t_0	t_{48}
0.25	4	0.22 (0.04)	0.15 (0.03)
0.5	4	0.56 (0.06)	0.51 (0.08)
1	4	0.97 (0.01)	0.95 (0.04)
2	4	2.38 (0.61)	1.84 (0.39)
4	4	3.62 (0.18)	3.13 (0.37)

† Duplicate samples taken at the beginning and at the end in each experiment have been pooled.

‡ Mean, with 1 SD in parentheses.

cadmium and density on age at first reproduction and on juvenile and adult survival probability rates were tested by sets of one-way Kruskal-Wallis ANOVA, because data of these variables were not normally distributed ($P < 0.05$, Kolmogorov-Smirnov tests). For each cadmium $\times$ density treatment, there was only one value for survival probability rates. Therefore, density and cadmium effects on survival probability rates were

assessed by pooling data among treatments. In cases for which treatment effects on life history responses were significant, nonparametric Tukey's multiple-range tests modified for unequal sample size (Zar 1996) were performed to test for pair-wise differences based on Kruskal-Wallis tests for subsets of data. To investigate influence of population density and cadmium on population growth rate responses, λ was regressed within each food level against ln-transformed population density and cadmium exposure levels. Linear, quadratic, and all possible interactions between the independent variables were considered to describe linear, antagonistic, and synergistic effects between population density and cadmium (Linke-Gamenick et al. 1999, Sibly 1999). Independent variables and their interactions were entered into the model in a step-wise fashion at a level of $P = 0.05$. Model fit was assessed by using the adjusted coefficient of determination r and by analysis of the distribution of residuals (Zar 1996). The overall significance of the regression and coefficients were determined by F and t tests, respectively (Zar 1996). Analyses were performed with the Statistical

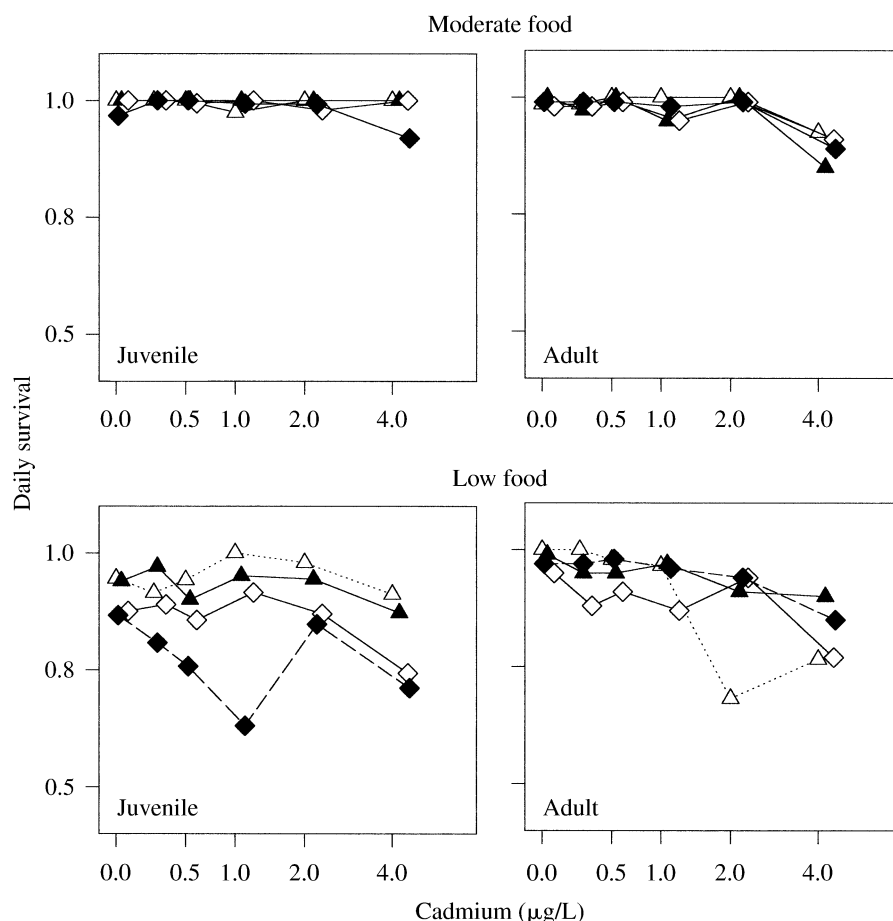


FIG. 3. Juvenile and adult survival rates (expressed as probability values per day) of *M. macleayi* females as a function of cadmium concentration and population density at moderate and low food levels. See Fig. 2 for symbol key.

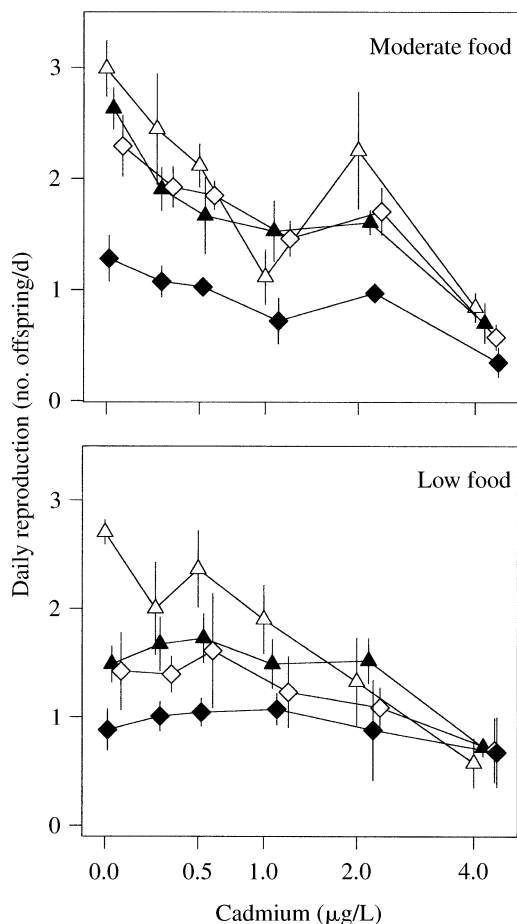


FIG. 4. Daily reproduction rates (mean $\pm$ 95% CL) of *M. macleayi* females as function of cadmium concentration and population density at moderate and low food levels. For clarity at some concentration levels mean values are shifted slightly to one side. See Fig. 2 for symbol key. ANOVA values for moderate food: Cd, $F_{5,83} = 21.0$, $P < 0.001$; D, $F_{3,83} = 10.2$, $P < 0.001$; Cd $\times$ D, $F_{15,83} = 1.57$, $P = 0.10$. ANOVA values for low food: Cd, $F_{5,62} = 4.0$, $P = 0.004$; D, $F_{3,62} = 4.3$, $P = 0.002$; Cd $\times$ D, $F_{15,62} = 2.4$, $P = 0.009$.

Package for Social Sciences (SPSS) 7.5.2 (1996) statistical package.

RESULTS

Chemical analysis

Measured concentrations of cadmium in old test solutions were not significantly different (t tests, $P > 0.05$) from freshly prepared solutions (Table 1), and actual concentrations of cadmium were in most cases within 10% of nominal concentrations. In all cases oxygen levels in old test solution were within 90% of saturation levels.

Survival responses

The effect of density and cadmium on survival responses varied with food availability (Fig. 3). At moderate food levels, juvenile survival rates were margin-

ally reduced at the highest population density ($\chi^2_3 = 8.8$, $P = 0.08$), whereas adult survival rates were significantly reduced at the highest cadmium concentration ($\chi^2_5 = 15.1$, $P = 0.01$). At low food levels, survival rates varied among densities and cadmium concentrations. More specifically, juvenile survival was significantly reduced with increasing population density ($\chi^2_3 = 14.7$, $P < 0.001$) and adult survival with increasing cadmium exposure ($\chi^2_5 = 12.7$; $P = 0.03$).

Reproductive responses

The effect of population density and cadmium on daily reproductive responses varied with food availability (Fig. 4). At moderate and low food levels, increasing exposure to cadmium and density had significant effects ($P < 0.05$) on daily reproduction rates. Significant interactions ($P < 0.05$) were only found at

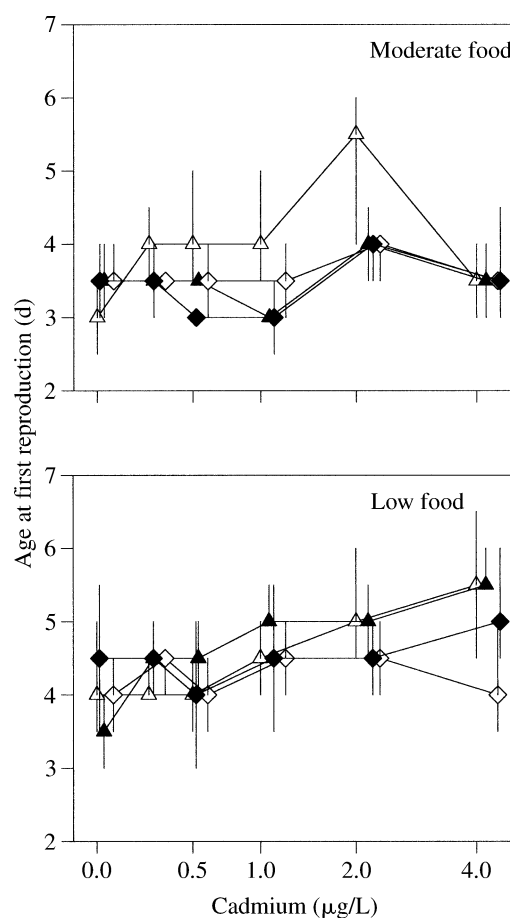


FIG. 5. Age at first reproduction (median $\pm$ 95% CL) of *M. macleayi* females as a function of cadmium concentration and population density at moderate and low food levels. ANOVA results are also shown. For clarity at some concentration levels mean values are shifted slightly to one side. See Fig. 2 for symbol key. ANOVA values for moderate food: Cd, $\chi^2_5 = 33.1$, $P < 0.001$; D, $\chi^2_3 = 4.1$, $P < 0.24$. ANOVA values for low food: Cd, $\chi^2_5 = 24.4$, $P < 0.001$; D, $\chi^2_3 = 7.4$, $P = 0.06$.

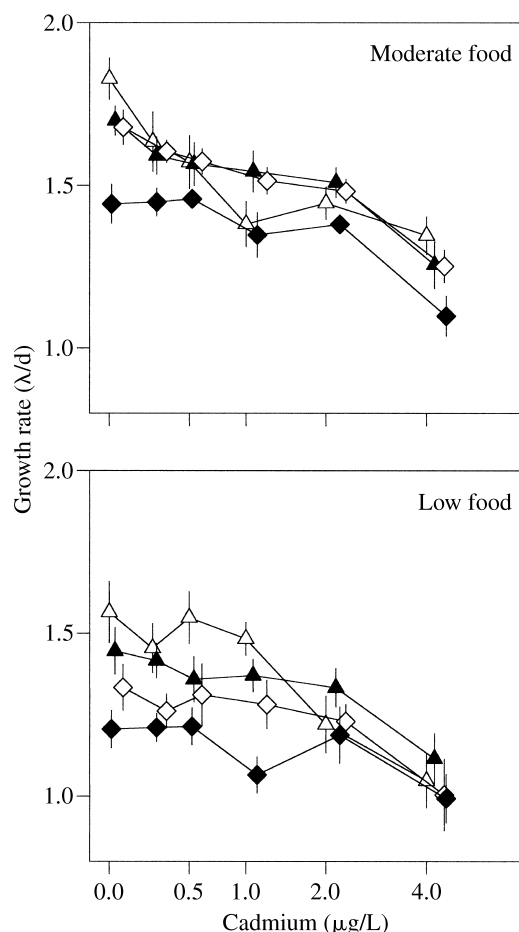


FIG. 6. Population growth rates, λ (mean $\pm$ 95% CL), of *M. macleanyi* females as a function of cadmium concentration and population density at moderate and low food levels. For clarity at some concentration levels mean values are shifted slightly to one side. See Fig. 2 for symbol key.

low food levels, indicating that the effect of cadmium varied among densities (Fig. 4). In particular, the effects of cadmium on reproduction were less apparent at population densities >25 animals/L.

At moderate and low food levels, exposure to cadmium significantly affected age at first reproduction. In contrast, an increase in population density had only a marginal effect on age at first reproduction at low food levels ($P = 0.06$; Fig. 5). Nevertheless, the effects of cadmium varied among densities. In general, age at first reproduction was delayed with increasing cadmium exposure levels at low but not at high population densities, indicating that an increase in population density was associated with reducing toxicant effect.

Population growth rate responses (λ)

The effects of density and cadmium on population growth rate responses (λ) varied with food availability (Fig. 6). At moderate food levels, exposure to cadmium had greater effects on λ than population density, where-

as at low food levels the opposite trend was observed. At both foods levels the effects of density on λ varied across cadmium exposure levels. At moderate food levels and cadmium concentrations of 0–2 $\mu\text{g/L}$, toxic effects reduced as density increased, but at higher cadmium concentrations the effects of density and cadmium were additive. At low food levels, the negative effects of cadmium on λ always reduced with increasing population density.

Population growth rate contours plotted against population density and cadmium concentrations, both $\ln$ -transformed, were convex at moderate and low food levels (Fig. 7A, C); indicating that increasing population densities ameliorated the effects of cadmium on population growth rates. Nevertheless, regression analysis indicated differences between food levels in the influence of population density and cadmium on population responses (Table 2). At moderate food levels, predicted contour lines (Fig. 7B) were convex at population densities of 125 animals/L and diagonal above it; indicating that density ameliorated toxic effects below a threshold density, above which the effects of density and cadmium were additive. As a result, at those densities close to the stable equilibrium population size (carrying capacity, contour $\lambda = 1$), predicted contour lines became diagonal, indicating additivity between density and toxicant stress. At low food levels, predicted contour lines (Fig. 7D) were convex until a threshold density, i.e., the carrying capacity (contour with $\lambda = 1$), was reached. Above the threshold density, contours were concave, indicating that antagonistic effects between density and cadmium toxicity increased with density. It is interesting to notice that at densities beyond the carrying capacity ($\lambda < 1$), population growth rates increased with exposure levels. Nevertheless, model predictions above the studied densities (>500 animals/L) are subject to extrapolation errors and should be interpreted with caution.

Decomposition of cadmium and density effects on λ

Decomposition of the two-stage model (Fig. 8) indicated that the contribution of survival rates, age at first reproduction, and daily reproduction to effects on λ varied among food levels and among densities. Table 3 showed which demographic variable had the greatest contribution to λ at a given food level, density, and cadmium concentration. At moderate food levels and low and moderate densities (25–250 animals/L), negative treatment effects on λ were derived from antagonistic contributions of daily reproduction rate and age at first reproduction, with the greatest contribution from daily reproduction rate. At the highest density, negative treatment effects on λ were mostly driven by negative contributions from daily reproductive rate. At low food levels and low densities (25 animals/L), daily reproduction rate, age at first reproduction, and adult survival rates contributed to λ changes, with the major contribution coming from daily reproductive rate. At

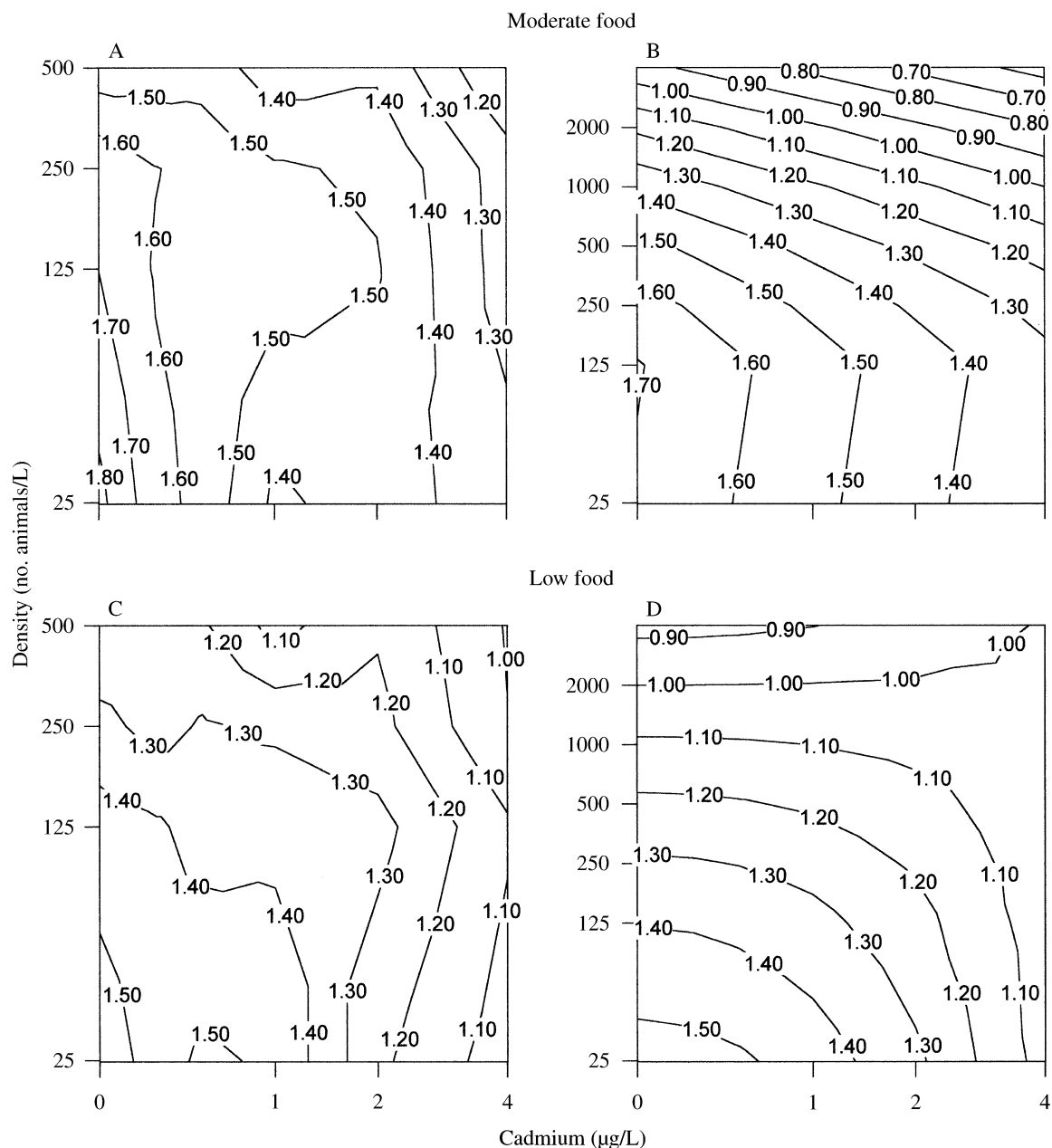


FIG. 7. Cadmium concentration–population density population growth rate (λ) contours at (A and B) moderate and (C and D) low food levels. Each line connects cadmium concentrations and population densities that result in a particular population growth rate. Predicted contours calculated from the best regression models obtained at moderate and low food levels (Table 2) are depicted in panels B and D. To include the $\lambda \leq 1$ contours, population densities above those studied here are also depicted.

densities of 125 and 250 animals/L, the four life history traits contributed to λ changes, with the greatest contribution from daily reproductive rate. At the highest density (500 animals/L), negative effects on λ arose from a trade-off between juvenile survival rate and daily reproduction rate. For both food levels at low and intermediate densities (25–250 animals/L), negative treatment effects on λ and on the contribution of life

history traits were mostly caused by cadmium, whereas at the highest density population density was the major contributor of reductions in λ .

DISCUSSION

According to Caswell (1996) and Levin et al. (1996), the extent to which population growth rate responds to toxicant exposure depends on (1) the severity of its

TABLE 2. Multiple regression models fitted to the concentration–density population growth rate responses (λ) of *M. macleayi* at moderate and low food levels.

Regression model	Regression coefficient				N	r^2
	a	b	c	d		
Moderate food level $\lambda = a + b[\text{Cd}] + c(D) + d(D)^2$	0.93	−0.23	0.38	−0.05	24	0.82
Low food level $\lambda = a + b[\text{Cd}]^2 + c(D)^2 + d[\text{Cd}]^2 (D)^2$	1.66	−0.23	−0.01	−0.004	24	0.88

Notes: For each food level the best models obtained in terms of r^2 are depicted. Prior to fitting the data to regression models, cadmium concentrations (X ; including controls) were $\ln(X + 1)$ transformed, and densities (D) were converted to natural logarithms. The former transformation allowed the inclusion of responses at 0 $\mu\text{g Cd/L}$. The terms $[\text{Cd}]$ and (D) indicate $\ln(\text{cadmium concentration} + 1)$ and $\ln(\text{population density})$, respectively, and N and r^2 are sample size and coefficient of determination of the regression. The regression models were highly significant ($P < 0.001$), regression coefficients were significantly different from zero ($P < 0.05$; t tests), and their residuals were normally distributed ($P > 0.05$; Shapiro-Wilk tests).

effect on individual life history traits, and (2) the sensitivity of λ to changes in each of its constituent life history traits. The results reported here show that effects of density and cadmium on life history traits with the greatest contribution to λ explained the differences in population dynamics observed between food levels. At moderate food levels and low densities, increasing population density ameliorated the retarding effects of cadmium on population growth rate (λ) responses. This is most probably explained by antagonistic effects between the contributions of age at first reproduction and daily reproduction rate on λ changes. At high densities, however, the effects of cadmium on daily reproduction rate increased with density, hence the effects of density and cadmium on λ were additive. At low food levels, the effects of density and cadmium on λ were always antagonistic; since where there were density-dependent effects on life history responses to cadmium, these were always antagonistic. Thus, the results reported for population responses at moderate food levels and high densities (>25 animals/L) and at low food levels support the hypothesis that at barely limiting food conditions, negative toxic effects on population dynamics are exacerbated by increasing density, due to the synergistic effects of density and toxicity on reproduction. Alternatively, at very limiting food conditions, when population density affects juvenile and adult survival, density-dependent effects are likely to compensate for toxicant-caused reductions in population growth rates since more food is available to the survivors, allowing production rates to increase (Sibly 1996). Moreover, our results agree with those of Kooijman and Metz (1984) who, modeling population responses of the cladoceran *Daphnia magna*, suggested that population responses to toxicants could be dramatically affected by the feeding status of the population, which in this present study is a function of density and food level.

Studying density-dependent effects of fluoranthene toxicity in the polychaete *Capitella*, Linke-Gamenick et al. (1999) found that, although juvenile and adult mortality were marginally affected by density, at low toxicant exposure levels increasing population density

ameliorated toxic effects on fecundity, on age at first reproduction, and on λ . At high toxicant exposure levels, however, increasing population density prevented worms from reproducing and hence exacerbated toxicant effects on λ . These results disagree with those obtained in this present study at moderate food levels. Nevertheless, in the study of Linke-Gamenick et al. (1999), juvenile survival decreased with increasing fluoranthene exposure levels, leaving more food per head among survivors, so that negative effects of increasing density on population growth rate were reduced by increasing fluoranthene toxicity. Thus the effects of density on population response to toxicants not only depends on the extent to which density affects mortality or production rates, but also on how the previous vital rates are affected by toxicants. Studying life history responses of *Daphnia magna* individuals to toxic chemicals, Barata and Baird (2000) showed that, for cadmium, concentration effects on production rates were almost an order of magnitude lower than those that affected adult or juvenile survival. Fluoranthene showed a different pattern, with juvenile survival during development and production being equally affected at concentrations that also had effects on adult survival. Thus, the contrasting dominant ecotoxicological modes of action between cadmium and fluoranthene, in terms of the balance between production and mortality effects, could possibly explain the discrepancies observed between our results and those from Linke-Gamenick et al. (1999).

Decomposition analysis showed that contributions of age at first reproduction on λ decreased dramatically with increasing density and/or with decreasing food availability. Alternatively, contributions of reproduction rates increased with density, regardless of the availability of food, whereas contributions of juvenile survival rates were especially important at high densities and very limiting food conditions. Grant (1998), simulating density-dependence effects on life table responses of the copepod *Eurytemora affinis* to diel drin using an age-classified model, showed that when populations were regulated by density, contributions of vi-

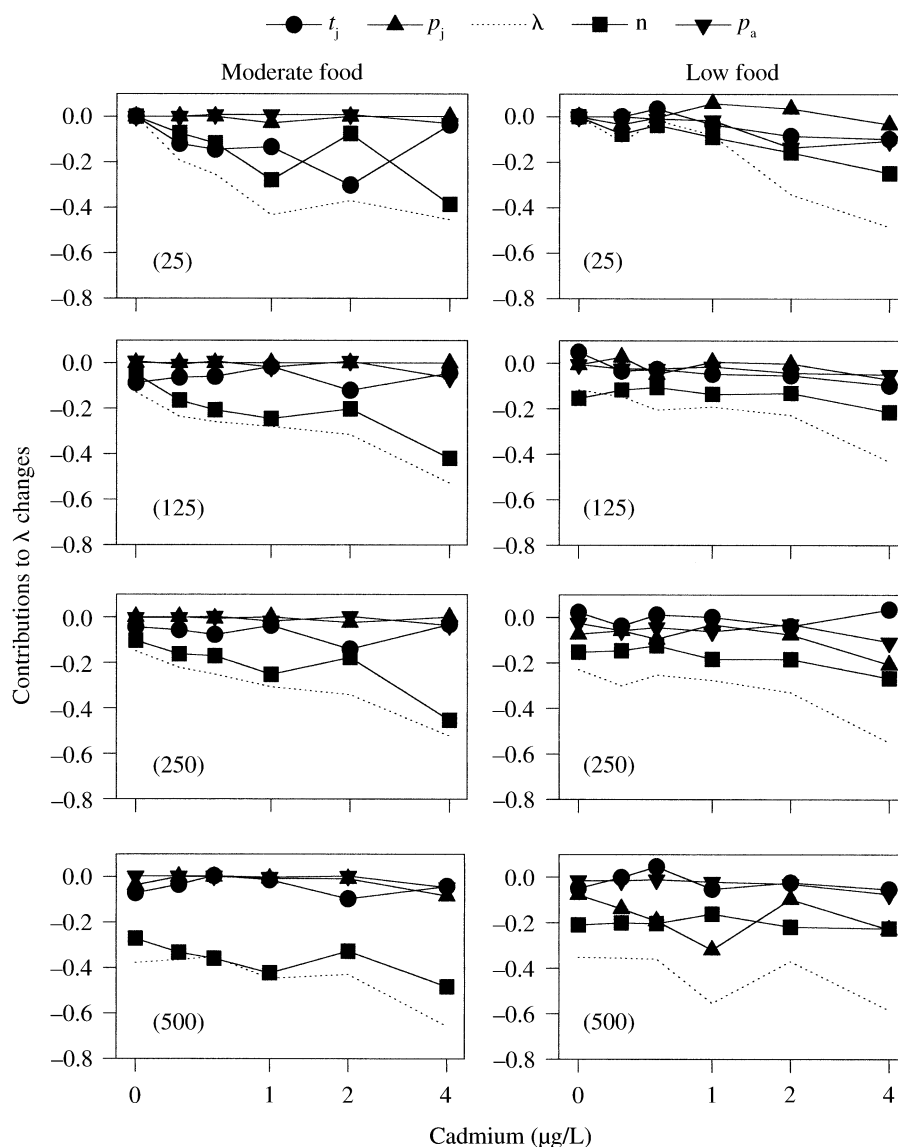


FIG. 8. Decomposition analysis for the two-stage model. Density-dependence effects on the contributions of daily reproduction rates (n), age at first reproduction (t_j), and juvenile (p_j) and adult (p_a) survival rates to differences in λ between the control (0 $\mu\text{g Cd/L}$) and cadmium exposure levels. The sums of the total contributions to λ (dotted lines) are also depicted. Population densities (no. individuals/L) appear in parentheses in each panel.

tal rates later in life were more important than in the density-independent analysis. Furthermore, Grant (1998) also showed that density-dependent effects on juvenile survival and fecundity contributed most to differences between treatments. In the present study, we used an age-independent demographic model, thus it was not possible to compare contributions of vital rates during the early and later stages of the life cycle. Nevertheless, we found that toxic effects on life history traits related to vital rates early in life, such as age at first reproduction, produced negligible effects on λ when populations were regulated by density. Conversely, effects on juvenile survival and reproduction rates had major effects on the population dynamics in den-

sity-regulated populations. Therefore, the findings reported in this present study are in agreement with Grant (1998).

To date, the performance of life table response experiments to evaluate population effects of chemicals have been conducted at low densities and high food levels (Kammega et al. 1996, Levin et al. 1996, Sibly 1996, Hansen et al. 1999). In almost all cases, these authors have reported that treatment effects early in life, such as variations in age at first reproduction or in the size of first broods, have major effects on population dynamics. Thus differences between the above-mentioned life table response studies and our results showed that the life history traits that make the greatest

TABLE 3. Demographic variables with greater contributions to λ across food, density, and cadmium treatments according to the decomposition analysis reported in Fig. 8.

Density (no. individuals/L)	Cadmium ($\mu\text{g/L}$)						
	0	0.2	0.5	1	2	4	0–4§
Moderate food levels							
25	none‡	t_j	$n \times t_j$	n	t_j	n	t_j (0.7†); n (0.9*)
125	t_j	n	n	n	$n \times t_j$	n	n (0.8*)
250	n	n	n	n	$n \times t_j$	n	n (0.9*)
500	n	n	n	n	n	n	n (0.8*)
Low food levels							
25	none‡	n	n	$n \times p_j$	$n \times p_a$	n	n (0.9*)
125	n	n	n	n	n	n	n (0.7†)
250	n	n	$n \times p_j$	n	n	$n \times p_j$	n (0.7†)
500	n	n	$n \times p_j$	p_j	n	$n \times p_j$	p_j (0.8*); n (0.7†)

Notes: Within each food and density level, correlation analyses between contributions of demographic variables and λ across cadmium concentrations (0–4) were performed. Those demographic variables that had the highest correlation coefficient were nominated as having the greatest contribution on λ . A nonparametric Spearman's rho correlation coefficient was used to account for departures of normality. The variables t_j , n , p_j , and p_a indicate age at first reproduction, daily reproduction, and juvenile and adult survival rates, respectively.

† $P < 0.1$; * $P < 0.05$.

‡ By definition, contributions of demographic traits on λ in control treatments are zero.

§ Values in parentheses indicate correlation coefficients and significance levels for the preceding trait.

contribution to treatment effects on λ are altered by the form of density dependence. In particular, our results and those of Grant (1998) showed that effects on age at first reproduction had little effect on population dynamics except at low densities and high food levels. This indicates that life table response experiments performed at high food conditions and at low densities may have limited ecological relevance since they are unable to identify the key life history traits contributing to population responses to stress at low food and high density levels.

Most populations in nature are likely to be at steady-state density (Calow et al. 1997). This means that the $\lambda = 1$ contour plotted in a plane with the axes being toxicant and density has a special ecological significance, because it indicates the stable equilibrium population size (carrying capacity; Sibly 1999). Thus the $\lambda = 1$ contour shows how carrying capacities change as stress levels increase. The results depicted in Fig. 7B, D showed that at limiting food concentrations, as stress levels increase, carrying capacity is reduced. However, at very limiting food levels, carrying capacity remained unchanged with increasing toxicant stress levels. Furthermore, at higher densities than the carrying capacity, contours with $\lambda < 1$ indicated that population growth rates increased with exposure to cadmium. These latter results have also been reported in previous experimental studies. Koivisto et al. (1992), studying the effect of food on population responses of four different cladoceran species to copper exposure, found that populations of *Daphnia galeata* would not grow at the lowest food level used, but exposure to copper increased population growth rates; yet at nonlimiting food levels, copper always reduced population growth rates. Postma et al. (1994), studying population responses of *Chironomus riparius* to cadmium at very high and low population densities, found that even very low concentrations of

cadmium were able to reduce population growth rates at low population densities. Conversely, at very high population density, exposure to cadmium ameliorated density dependence effects; population growth increased at moderate levels of cadmium and decreased again at high cadmium concentrations, but did not differ from the control. On the contrary, van Leeuwen et al. (1985) using a logistic demographic model found that the carrying capacity of *Daphnia* populations decreased linearly with increasing cadmium exposure levels. The food supply used in the van Leeuwen et al. (1985) study was not limiting, at least at low population densities, and their results were similar to those obtained in the present study at moderate food levels. Therefore, results from this present study and those reported in previous experimental work showed that population responses to toxicants at the stable equilibrium size are altered by the feeding status of the population. In a simulation study, Grant (1998) found that when juvenile survival was affected by population density, the impact of toxicants on equilibrium population size was less marked than was their impact on population growth rate in density-independent regulated populations. Conversely, when fecundity was affected by population density, toxicant effects on equilibrium population size were more marked than toxicant effects on population growth rate in density-independent regulated populations. Our results also agree with the predictions achieved by Grant (1998) because, at moderate food levels, increasing population density exacerbated toxicant effects on λ throughout population density effects in fecundity, whereas at low food levels juvenile survival was dramatically affected by population density, and increasing population density ameliorated toxicant effects on λ .

CONCLUSIONS

In conclusion, our results showed that the demographic analysis of toxicant effects on population dy-

namics can be strongly influenced by food supply and population density through effects on mortality and production rates. The demographic model used in this study makes a number of simplifying assumptions. For example, it was assumed that mortality and birth rates were independent of age. Our results and those of previous studies have showed that the assumptions that birth and mortality rates do not change with age may also be reasonable for organisms that grow after the age of first breeding (Sibly 1996, Linke-Gamenick et al. 1999, Sibly et al. 2000). Therefore this approach can be seen as an advance over previous life table response studies, since the experimental design allows incorporation of density-dependent effects on population response to toxic chemicals when only production and/or mortality rates were affected by density.

It was notable that the effects of toxic stress on steady-state population size was compensatory when mortality rates were affected by population density, whereas the opposite trend was observed only when production rates were affected. This means that ecological compensation due to density-dependent effects on juvenile mortality will prevent populations at the steady-state equilibrium size from becoming extinct though toxic stress, particularly in oligotrophic conditions. Alternatively under eutrophic conditions, when production rates (rather than mortality) dominate population dynamics, exposure to toxic chemicals may increase the probability of extinction of sensitive species populations. Nevertheless, field populations, including those of *Daphnia* species, are usually in a dynamic equilibrium, experiencing periods of fast growth with low mortality rates and high birth rates followed by periods with negative growth due to high mortality rates or low birth rates (McCauley and Murdoch 1987). Therefore future work needs to be focused on developing risk assessment procedures on the basis of demographic models that integrate risks of extinction associated with density dependence and stress effects in growing and decreasing populations (Calow et al. 1997).

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